

Computational Techniques in Civil Engineering

TU, IOE
B.E. (Civil), Year/Part: IV/II

Lecture Notes on Introduction to Computational Techniques
and Application of Finite Difference Method (FDM) to Water
Resources Engineering

Prepared by
Dr. K. N. Dulal
Year: 2013

Chapters of syllabus included in the notes

	Page no.
Chapter 1: Introduction to Numerical computation	1
Chapter 2: Solution of Equations	6
Chapter 5: Finite Difference Method (FDM)	11
Chapter 6: Method of Characteristics (MOC)	23
Chapter 7: Simulation of Groundwater Flow	28

Chapter 1: Introduction to Numerical computation

1.1 Introduction

Physical processes in the various field of Civil Engineering are described by differential equations. These equations are to be solved to obtain the value of variables. The analytical solution is possible for simplified governing equations. However, for complex problems, it is difficult to get analytical solutions. Therefore, solutions of the equations are obtained by approximate method using computer. This is known as Computational Technique. In this approach, Numerical methods are applied for the solution.

Terms

Domain

A domain describes a finite or infinite continuous region i.e. geometry of the system.

Boundary conditions

The boundary conditions (BC) represent the given or known conditions on some parts of a domain.

Procedure of constructing a numerical simulation

a. Conceptualization of Physical phenomenon

- Various laws describing physics, simplification and extraction of important physics

b. Mathematical model

- Formulation of governing equations: Ordinary differential equations (ODE) or Partial differential equations (PDE)

c. Domain discretization

- Representing domain by mesh/grid, cell, node or particle

d. Numerical algorithms

- Initial condition, boundary conditions
- Numerical discretization: converting continuous ODE/PDE to discrete form (algebraic form or PDE to ODE)
- Solution steps for resulting equations

e. Coding and implementation

f. Numerical simulation

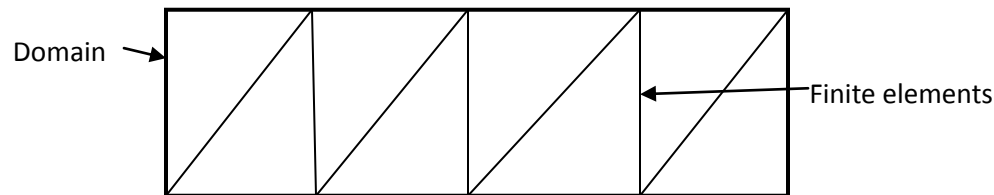
1.2 Introduction to different types of numerical methods

I. Finite element method (FEM)

In Finite element method, the domain is divided into a finite number of small, interconnected elements/mesh called finite elements. Nodes are assigned to each element. Over each finite element, the unknown variables are approximated using known functions, known as approximating

functions/interpolation functions. These functions are defined in terms of the field variable at specified nodes. The unknowns at a node are magnitude of field variables and its derivative. The governing equations are integrated over each finite element and the solution is summed up over each element. A set of finite linear equations are obtained in terms of unknown parameters over each element and these equations are solved by linear algebraic techniques. Once the unknowns at nodes are found, the field variables at any point can be found by interpolation functions.

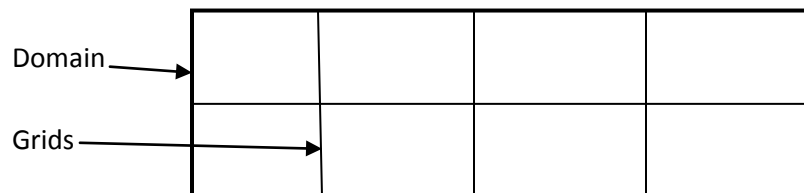
It is more versatile method and useful for irregular geometry, but it cannot model discontinuous structures.



II. Finite difference method (FDM)

In Finite difference method, the domain is represented by an array of regularly spaced grid points (rows and columns). The differential equation is converted to algebraic difference equation at a grid point. Hence, the method approximates the governing equation pointwise. The value of a function at a point is related to value of nearby points.

The method is simple and easy to implement, but it is difficult to apply for irregular geometry.



Comparison of FEM/FDM

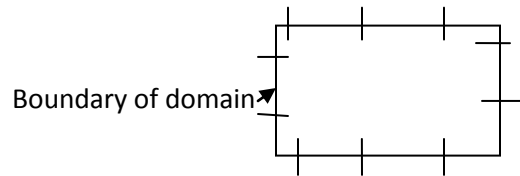
Main difference

In FDM, field variable is computed at specified points, but in FEM, the variation of field variable is integral part of the problem formulation.

The FDM models the differential equation only, but FEM models the entire domain of the problem.

Similarities: The integration points in FDM is analogous to the nodes in FEM as the variable of interest is evaluated at such point. In both methods, differential equation is converted to algebraic equation.

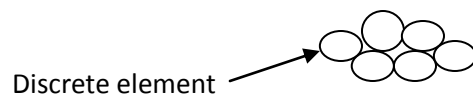
III. Boundary element method (BEM)



In Boundary element method, only the boundary of the solution domain is discretized into elements. The idea of boundary element methods is that we can approximate the solution of a differential equation by looking at the solution on the boundary and then use that information to get the solution inside the domain. This method reduces the preprocessing time and the dimensionality. In BEM, the differential equation is converted into an integral equation on the boundary of the domain (boundary integral). Integration over the boundary surface is carried out over a boundary element and the contributions of all elements describing the boundary are added. Once the solution at the boundary is obtained, then the solution inside the domain is obtained.

IV. Discrete Element method (DEM)

In Discrete Element method, a granular material is modelled as an assembly of separate, discrete particles. These particles may have different shapes and properties. This method is used to compute the stresses and displacements in a volume containing a large number of particles such as grains of sand. In DEM simulation, all particles are oriented spatially and an initial velocity is assigned. The forces which act on each particle are computed from the initial data and the relevant physical laws and contact models. All forces are added up to find the total force acting on each particle. An integration method is employed to compute the change in the position and the velocity of each particle during a certain time step from Newton's laws of motion. Then, the new positions are used to compute the forces during the next step, and this loop is repeated until the simulation ends. The method is computationally intensive.



V. Smoothed particle Hydrodynamics (SPH)

Grid based methods limitations are that mesh generation for complex geometry possess difficulties; they are not suitable for free surface, deformable boundary, discontinuities, and moving interface.

Smoothed particle Hydrodynamics (SPH) is a mesh free method based on Lagrangian concept of Fluid dynamics. It is a particle method, in which the system/domain is represented by a set of arbitrarily

distributed particles. No connectivity of particles is needed. The physical quantity of any particle is taken by computing average over neighbouring particles (smoothing).

Steps

- Integral representation of a function
- Integral representation of a function converted to particle approximation
- Particle approximation performed at every time step
- Particle approximation performed to all terms related to field functions in partial differential equations to produce a set of ODES in discretized form with respect to time only.
- Solving ODES

1.3 History of numerical computations in Civil Engineering

FDM history

- Oldest method
- Foundation of FDM: Fundamental paper by Courant, Friedrichs and Lewy (1928) on the solutions of the problems of mathematical physics
- A number of schemes developed after 1950s, which were applied in the field of Civil engineering
- Wide applications due to the emergence of high speed computer after 1980

FEM history

- Basic idea originated from the framed structures like truss, aircraft structural analysis and flow network analysis
- Aircraft engineers in 1940s developed flexibility method (unknown force, known displacements)
- FEM: stems concept from displacement method (known force, unknown displacements)
- 1940s: Courant developed methods for solving torsion problems
- 1955: Argyris published a book on Structural analysis which provides foundation for FEM
- 1956: Turner and others developed stiffness matrices for truss, beam and other elements
- The term FEM was first used by Clough in 1960 in the context of plane strain analysis
- 1960 and 1970s: FEM applied to plate bending, shell bending, pressure vessels, 3d problems in structural analysis, fluid flow and heat transfer
- 1980: Zienkiewicz and Cheung published first book on FEM
- 1980: Graphical and computational development
- 1990: emergence of low cost, powerful computers, possibility of analysis of large structures

BEM history

- The term was first used in the 1977 paper "Boundary element methods for potential problems" by Brebbia and Dominguez.
- Ideas obtained from famous Mathematicians like Laplace, Gauss, Fredholm, Betti.

DEM history

- First applied to rock mechanics and soils by Cundall and Strack in 1979

SPH history

- Introduced in 1977 to solve astrophysical problems
- Applied to Fluid dynamics related area in the early phase

Softwares/applications

FEM softwares

Structure: STAAD, ANSYS, NISA, RAM

Geotech: STAAD Foundation/RAM Foundation, CRISP

Fluid Mechanics: ANSYS CFX, FEFLOW, FEMWATER

FDM softwares: HEC-RAS, MikeShe

Applications of different methods in civil engineering

FEM Applications

Structural analysis

- Computation of deflection, stress, strain, force, displacements, energy

Fluid flow analysis

- Computation of velocity, pressure

Groundwater flow analysis

FDM applications

- Hydraulics, surface water and groundwater hydrology

BEM applications

- Fluid flow analysis, structural analysis

DEM applications

- Mechanical behaviour of soil, rock and concrete

SPH applications

- Fluid flow analysis

Chapter 2: Solution of Equations

2.1 Solution of linear equations

$$a_{11} X_1 + a_{12} X_2 + \dots + a_{1n} X_n = a_{1,n+1}$$

$$a_{21} X_1 + a_{22} X_2 + \dots + a_{2n} X_n = a_{2,n+1}$$

.....

.....

$$a_{n1} X_1 + a_{n2} X_2 + \dots + a_{nn} X_n = a_{n,n+1}$$

Matrix form

$$AX=b$$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & \dots & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & \dots & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & \dots & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ \dots \\ \dots \\ X_n \end{bmatrix} = \begin{bmatrix} a_{1,n+1} \\ a_{2,n+1} \\ \dots \\ \dots \\ a_{n,n+1} \end{bmatrix}$$

Solution technique

a. Gaussian elimination

By row, column operation, convert the matrix A to upper triangular matrix and perform backward substitution.

$$\begin{bmatrix} a_{11} & a_{12} & \dots & \dots & \dots & a_{1n} \\ 0 & a_{22} & \dots & \dots & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \dots & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ \dots \\ \dots \\ X_n \end{bmatrix} = \begin{bmatrix} a_{1,n+1} \\ a_{2,n+1} \\ \dots \\ \dots \\ a_{n,n+1} \end{bmatrix}$$

b. Gaussian-Jordan elimination

By row, column operation, convert the matrix A to identity matrix (Make all diagonal elements 1 and non-diagonal element 0).

$$\begin{bmatrix} 1 & 0 & \dots & \dots & \dots & 0 \\ 0 & 1 & \dots & \dots & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \dots & \dots & 1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ \dots \\ \dots \\ X_n \end{bmatrix} = \begin{bmatrix} a_{1,n+1} \\ a_{2,n+1} \\ \dots \\ \dots \\ a_{n,n+1} \end{bmatrix}$$

Inversion using Gaussian-Jordan

$$A^{-1}A=1$$

$$A^{-1} \begin{bmatrix} a_{11} & a_{12} & \dots & \dots & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & \dots & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & \dots & \dots & a_{nn} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \dots & \dots & \dots & 0 \\ 0 & 1 & \dots & \dots & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \dots & \dots & 1 \end{bmatrix}$$

By row, columns operations on both sides, convert matrix A into identity matrix. The resulting matrix on RHS will be the inversion of matrix A.

c. Gauss Seidel iteration

$$a_{11} X_1 + a_{12} X_2 + \dots + a_{1n} X_n = a_{1,n+1}$$

$$a_{21} X_1 + a_{22} X_2 + \dots + a_{2n} X_n = a_{2,n+1}$$

.....

.....

$$a_{n1} X_1 + a_{n2} X_2 + \dots + a_{nn} X_n = a_{n,n+1}$$

Rewriting above equations

$$X_1 = \frac{1}{a_{11}} \{a_{1,n+1} - \sum_{j=2}^n a_{1j} X_j\}$$

$$X_2 = \frac{1}{a_{22}} \{a_{2,n+1} - \sum_{j=1}^n a_{2j} X_j\} \text{ for } j \neq 2$$

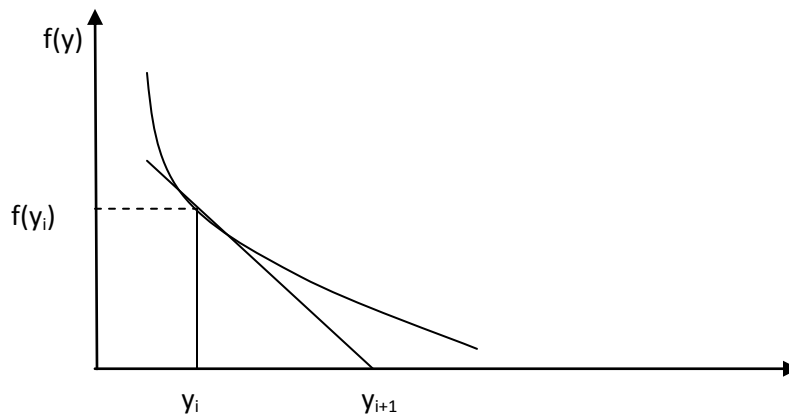
$$X_n = \frac{1}{a_{nn}} \{a_{n,n+1} - \sum_{j=1}^n a_{nj} X_j\} \text{ for } j \neq n$$

Solving steps

- Assign initial values to $X_1, X_2, \dots, X_n$.
- Substitute the initial values to RHS of eq. (1) to obtain new X_1 . Substitute new X_1 and other old values to eq. (II) to obtain new X_2 . Similarly, compute other new values by substituting recently computed new values and old values.
- Continue the iterations until the new value and old value does not differ significantly.

2.2 Iteration techniques

a. Newton-Raphson method for solving non-linear equation containing only one dependent variable



Slope of function at i

$$\left(\frac{df}{dy}\right)_i = \frac{0 - f(y_i)}{y_{i+1} - y_i}$$

$$y_{i+1} = y_i - \frac{f(y_i)}{f'(y_i)}$$

$$\text{Where } f'(y_i) = \left(\frac{df}{dy}\right)_i$$

Knowing y_i , compute y_{i+1} .

y_i for next step = y_{i+1} of previous step

Continue computation until two values of y_i does not differ significantly.

b. Gauss-Seidel method as described above

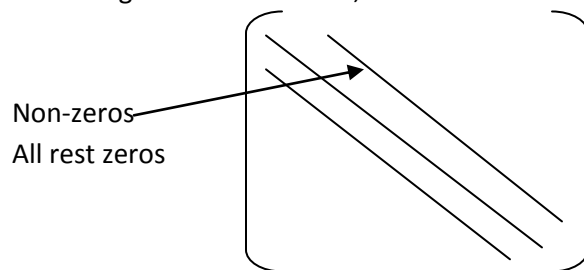
2.3 Sparse and banded matrix

Sparse matrix

A matrix whose most of the elements are zero is called sparse matrix. Such matrix occurs in many applications while solving PDF by numerical methods such as FDM and FEM. Storing only the non-zero entries and their location saves huge computer memory.

Band matrix

A matrix whose non-zero elements occur close to main diagonal within a small bandwidth and all the remaining elements are zero, is called band matrix. Such matrix occurs in FDM and FEM applications.



Data storage and memory optimization

The banded matrix has elements defined on its main diagonal, on k_l sub-diagonals (above main diagonals) and on k_u superdiagonals (below main diagonal). In solving problems containing such matrix, only the non-zero elements are stored. Various storage formats are available. The most common method is to store m -by- n band matrix in a two-dimensional array with k_l+k_u+1 rows and n columns. Columns of the matrix are stored in corresponding columns of the array, and diagonals of the matrix are stored in rows of the array.

2.3 Conjugate-gradient method

The conjugate-gradient method is an iterative technique for solving large systems of linear equations $Ax = b$ when the coefficient matrix A is symmetric ($A^T=A$) and positive definite ($x^T Ax > 0$).

The problem is to minimize $f(y) = \frac{1}{2} x^T A x - b^T x$. This minimization is equivalent to solving $AX=b$.

Conjugate means two unequal vectors orthogonal to each other. Conjugate set is generated by the

gradient vectors. Search direction is constructed by conjugation of residuals. Each residual is orthogonal to the previous search direction as well as to the previous residuals.

$\nabla f(x)$ = Gradient of function at X

Residual $r = -\nabla f(x)$

At $x = x_0$, gradient of f is $Ax_0 - b$, Conjugate vector = $b - Ax_0$, Residue at i , $r_i = b - Ax_i$

Move in direction of r_i

Algorithm

$r(0) = b - Ax(0)$

$d(0) = r(0)$

$$\alpha(i) = \frac{r_i^T r_i}{d_i^T A d_i}$$

Next optimal location

$$x(i+1) = x(i) + \alpha(i)d(i)$$

$$r(i+1) = r(i) - \alpha(i)A d(i)$$

If $r(i+1)$ is small, exit . Solution: $x(i+1)$

Else

$$\beta(i+1) = \frac{r_{i+1}^T r_{i+1}}{r_i^T r_i}$$

$$d(i+1) = r(i+1) + \beta(i+1)d(i)$$

2.4 Fourier transform

In general, curve fitting for a set of data points can be done by a linear combination of polynomial functions, with base functions $1, x, x^2, \dots, x^m$. The unknown coefficients can be obtained by solving the associated linear simultaneous equations. Alternatively, trigonometric functions can also be used as based functions, in which the unknown coefficients can be found without requiring solving the simultaneous linear equations. The Fourier series is such a technique to represent the functional relationship of a variable in trigonometric form, i.e. by using sum of infinite number of sine and cosine terms.

The Fourier series representation for a variable is written as

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega t) + b_n \sin(n\omega t)] \quad (a)$$

The Fourier transform of the original signal in time domain, $f(t)$ to frequency domain is

$$f(j\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

Where $\omega = 2\pi f$, $f = 1/T$

$f(j\omega)$ = Fourier integral or Fourier transform in complex form

$$e^{-jx} = \cos x + j \sin x, e^{jx} = \cos x - j \sin x$$

f = frequency (cycle/sec), T = Period(S)

The transformation of discrete data between time and frequency is accomplished by using Fourier transform.

Discrete Fourier transform (DFT)

If N sample data is available with T time interval, then discrete form of Fourier transform is used, which is given by

Data: $f(0), f(1), \dots, f(k), \dots, f(N-1)$

$$f(j\omega) = \int_0^{(N-1)T} f(t) e^{-j\omega t} dt$$

$$f(j\omega) = \sum_{k=0}^{(N-1)T} f(k) e^{-j\omega kT}$$

This equation represents the Discrete Fourier Transform (DFT).

n = sample number, T = time interval, $f(j\omega) = f(n)$

$$f(n) = \sum_{k=0}^{(N-1)T} f(k) e^{-j\omega knT}$$

For one cycle: $\omega = \frac{2\pi}{NT}$

$$f(n) = \sum_{k=0}^{(N-1)T} f(k) e^{-j\frac{2\pi kn}{N}} \text{ and } n = 0:N-1$$

Fast Fourier Transform (FFT)

Computation of DFT is slow as a large number of multiplications and additions are required for the calculations. For N-point DFT there will be N^2 multiplications and $N(N-1)$ additions. Highly efficient computer algorithms for estimating Discrete Fourier Transforms have been developed since the mid-60's. These are known as Fast Fourier Transform (FFT) algorithms.

Rewriting DFT: $f(n) = \sum_{k=0}^{(N-1)T} f(k) e^{-j\frac{2\pi kn}{N}}$ as

$$f(n) = \sum_{k=0}^{(N-1)T} f(k) W_N^{nk} \text{ where } W_N = e^{-j\frac{2\pi}{N}}$$

Same values of W_N^{nk} are calculated many times as the computation proceeds, W_N^{nk} is a periodic function with N distinct values (calculate only once).

Decimation in time

The single summation can be split over N samples into 2 summations, each with N/2 samples, one for k even and the other for k odd.

$K = 2m$ (even), $k = 2m+1$ (odd)

$$f(n) = \sum_{m=0}^{\frac{N}{2}-1} f(2m) W_N^{2mn} + \sum_{m=0}^{\frac{N}{2}-1} f(2m+1) W_N^{(2m+1)n}$$

$$W_N^{2mn} = e^{-j\frac{2\pi(2mn)}{N}} = e^{-j\frac{2\pi mn}{N/2}} = W_{N/2}^{mn}$$

$$f(n) = \sum_{m=0}^{\frac{N}{2}-1} f(2m) W_{N/2}^{mn} + W_N^m \sum_{m=0}^{\frac{N}{2}-1} f(2m+1) W_{N/2}^{mn}$$

$$f(n) = G(n) + W_N^m H(n)$$

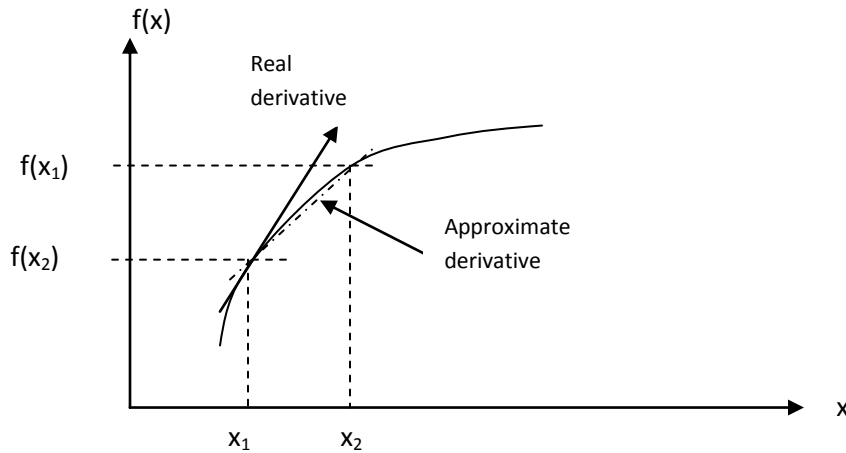
The n point DFT can be obtained from two N/2 point transforms, one on even input $G(n)$ and the other on odd input, $H(n)$.

The FFT is computed by dividing up, or decimating, the sample sequence into sub-sequences until only 2-point DFT's remain. Since it is the input, or time, samples which are divided up, this algorithm is known as the decimation in-time (DIT) algorithm.

Chapter 5: Finite Difference Method (FDM)

5.1 Introduction

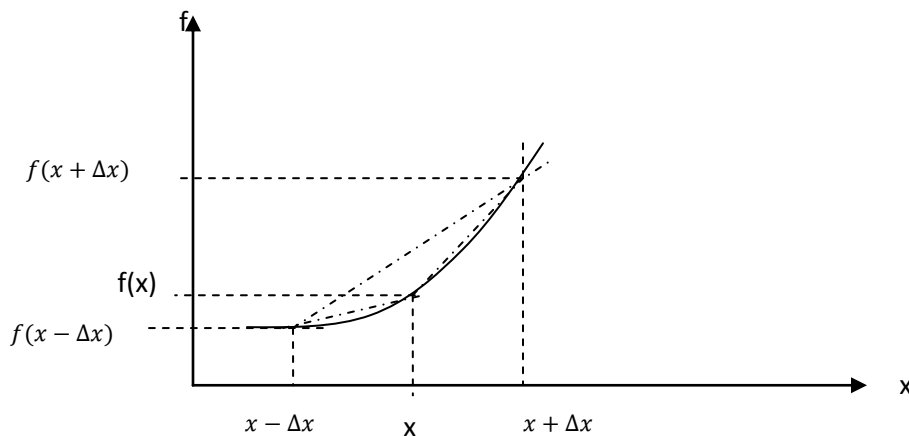
FDM is a technique of solving PDES by replacing them with difference equations over a small step.



Real derivative is represented by tangent to a point on a curve. For example, the approximate derivative can be represented as

$$\frac{df}{dx} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

Finite difference equations (FDE) are obtained by using Taylor's series approximation to PDE.



Forward difference approximation of derivative

Taylor's series expansion of $f(x)$ at $x + \Delta x$ is

$$f(x + \Delta x) = f(x) + \Delta x \frac{\partial f}{\partial x} + \frac{\Delta x^2}{2!} \frac{\partial^2 f}{\partial x^2} + \frac{\Delta x^3}{3!} \frac{\partial^3 f}{\partial x^3} + \dots \dots \quad (a)$$

Neglecting second and higher order terms

$$\frac{\partial f}{\partial x} = \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

Dropping the higher order terms is known as truncation.

Backward difference approximation of derivative

Taylor's series expansion of $f(x)$ at $x - \Delta x$ is

$$f(x - \Delta x) = f(x) - \Delta x \frac{\partial f}{\partial x} + \frac{\Delta x^2}{2!} \frac{\partial^2 f}{\partial x^2} - \frac{\Delta x^3}{3!} \frac{\partial^3 f}{\partial x^3} + \dots \dots \quad (b)$$

Neglecting second and higher order terms

$$\frac{\partial f}{\partial x} = \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

Central difference approximation of derivative

Subtracting eq. (b) from eq. (a), and neglecting third and higher order terms

$$f(x + \Delta x) - f(x - \Delta x) = 2\Delta x \frac{\partial f}{\partial x}$$

$$\frac{\partial f}{\partial x} = \frac{f(x + \Delta x) - f(x - \Delta x)}{2\Delta x}$$

Steps in FDM

- I. Discretization of domain: Replacing the continuous spatial and temporal domain by grid points or cells and time levels
- II. Discretization of PDE: Replacing PDE by a set of algebraic equations known as Finite difference equations
- III. Specification of solution algorithm: Step by step procedure for solving FDE at each point/grid.

Consistency, convergence and stability

If continuous PDES (real) are obtained from FDEs as Δx and Δt approaches to zero, then the resulting solution is said to be consistent. (Consistency related to equations). Taylor's series expansion is applicable for consistency analysis.

If solution of FDM approaches the true solution as Δx and Δt approaches to zero, then the resulting solution is said to be convergent. (Convergence related to solution of equations).

If the round-off and truncation errors do not accumulate to cause the solution to diverge, then the solution is said to be stable. Fourier analysis is performed for stability analysis.

Order of accuracy of schemes

Order of accuracy of represents how well the solution of finite difference method approximates the real solution of differential equations. It is an extension of consistency concept.

Consider the forward scheme.

Taylor's series expansion of $f(x)$ at $x + \Delta x$ is

$$f(x + \Delta x) = f(x) + \Delta x \frac{\partial f}{\partial x} + \frac{\Delta x^2}{2!} \frac{\partial^2 f}{\partial x^2} + \frac{\Delta x^3}{3!} \frac{\partial^3 f}{\partial x^3} + \dots \dots$$

Neglecting second and higher order terms

$$\frac{f(x+\Delta x)-f(x)}{\Delta x} - \frac{\partial f}{\partial x} = \frac{\Delta x}{2} \frac{\partial^2 f}{\partial x^2} + \frac{\Delta x^2}{6} \frac{\partial^3 f}{\partial x^3} + \dots$$

RHS = truncation error

The lowest power of Δx in the truncation error determines the order of accuracy. This scheme is first order accurate. As Δx tends to zero, the error in the higher order scheme tends to zero more rapidly than lower order scheme.

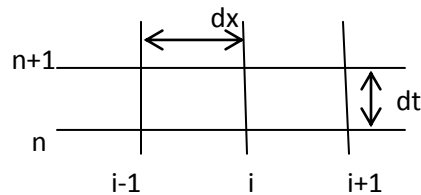
5.2 Explicit and implicit schemes

If the solution at $t = n+1$ is obtained from the conditions at $t = n$, the scheme is known as explicit scheme. In such scheme, the equations are solved sequentially. The explicit scheme is simpler to implement, but it can be unstable. It is also convenient because results are given at grid points and it can treat slightly varying channel geometry from section to section. It is less efficient than implicit method and hence not suitable for simulation over long time period.

If the solution at $t = n+1$ is obtained from the conditions at $t = n$ and $n+1$, the scheme is known as implicit scheme. In such scheme, the equations are solved simultaneously. Implicit scheme is mathematically more complicated. The method is stable for large computation steps and hence works much faster than explicit scheme. The method can handle channel geometry varying significantly from one section to the next.

Discretization

finding $\left(\frac{\partial f}{\partial x}\right)_i^{n+1}$



Explicit FDEs

Forward difference: $\frac{\partial f}{\partial x} = \frac{f_{i+1}^n - f_i^n}{\Delta X}$

Backward difference: $\frac{\partial f}{\partial x} = \frac{f_i^n - f_{i-1}^n}{\Delta X}$

Central difference: $\frac{\partial f}{\partial x} = \frac{f_{i+1}^n - f_{i-1}^n}{2\Delta X}$

Implicit FDEs

Forward difference: $\frac{\partial f}{\partial x} = \frac{f_{i+1}^{n+1} - f_i^{n+1}}{\Delta X}$

Backward difference: $\frac{\partial f}{\partial x} = \frac{f_i^{n+1} - f_{i-1}^{n+1}}{\Delta X}$

Central difference: $\frac{\partial f}{\partial x} = \frac{f_{i+1}^{n+1} - f_{i-1}^{n+1}}{2\Delta X}$

Second order derivative: Example of discretization

$$\begin{aligned}\frac{\partial^2 f}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{1}{\Delta x} \left[\left(\frac{\partial f}{\partial x} \right)_{i+1}^n - \left(\frac{\partial f}{\partial x} \right)_i^n \right] = \frac{1}{\Delta x} \left[\frac{f_{i+1}^n - f_i^n}{\Delta X} - \frac{f_i^n - f_{i-1}^n}{\Delta X} \right] \\ &= \frac{1}{\Delta x^2} (f_{i+1}^n - 2f_i^n + f_{i-1}^n)\end{aligned}$$

5.3 Unsteady non-uniform flow equation in Open channel flow: Saint Venant Equations

The equation of unsteady non-uniform flow in 1D is known as Saint Venant Equations. These equations are based on continuity and momentum principle.

Assumptions

- The channel is prismatic.
- The flow is 1D, i.e. the velocity is uniform in a cross-section.
- Hydrostatic pressure prevails and vertical accelerations are negligible.
- Bottom slope of the channel is small.
- Manning's equation is used to describe resistance effects (to evaluate friction slope).
- The fluid is incompressible.

Full equations

Continuity equation: Using the principle inflow-outflow = rate of change of storage

$$\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = q$$

Momentum equation: Based on Newton's second law

Net force = rate of change of momentum

$$\frac{\partial Q}{\partial t} + \frac{\partial (\beta Q^2 / A)}{\partial x} + gA \left(\frac{\partial y}{\partial x} + S_f - S_0 + S_e \right) + W_f b - \beta q v_x = 0$$

Q = discharge at x, A = Cross-sectional area at x, q = lateral flow per unit width, β = momentum coefficient, S_f = energy slope, S_0 = bed slope, S_e = eddy loss slope, W_f = wind shear factor, b = top width, v_x = velocity of lateral flow in x direction

Simplified and most widely used form of St. Vt. equations

The most common form of equation (taking $\beta = 1$, Neglecting eddy loss, wind shear and lateral flow)

Continuity: $\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = 0$ (I)

Momentum: $\frac{\partial Q}{\partial t} + \frac{\partial(Q^2/A)}{\partial x} + gA \frac{\partial y}{\partial x} - gA(S_0 - S_f) = 0$ (II)

Terms in momentum equation

Effect of inertia force: first and second terms

First term: local acceleration

Second term: convective acceleration

Third term: pressure force

Fourth term: gravity force

Fifth term: friction force

In the Saint Venant equations, all the flow variables are functions of both time and distance along the channel. The unknowns are Q and y in conservative forms of equations, and V and y in non-conservation form. The other flow variables, such as the area, A, and the friction slope, S_f , can be expressed in terms of Q and y. The independent variables are time, t, and distance along the channel, x. An initial condition and two boundary conditions are needed to solve the Saint Venant equations. The initial condition is described by the variation of the unknowns, Q and y (or V and y), along the channel at time zero.

Simplifications of St. Venant equations

a. Dynamic wave model: Full momentum equation is considered in Dynamic wave model.

Equations (I) and (II) represent equations for dynamic wave model.

Application

- Flood wave in a wide river
- For rivers in which flood is modified by developments and structures

b. Diffusive wave model

Neglecting first two terms (inertia) of momentum eq., diffusive wave eq. is obtained. Acceleration terms are small compared to water surface slope for slow rising of flood wave, which can be neglected.

Continuity: $\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = 0$

Momentum: $\frac{\partial y}{\partial x} = S_0 - S_f$ (III)

• Application

- To represent backwater effect, steep slope channel and flow influenced by bed roughness

c. Kinematic wave model

Neglecting first three terms of momentum eq., kinematic wave equation is obtained.

Continuity: $\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = q$ (a)

Momentum: $S_0 = S_f$ (b)

In this model, gravity force and friction force balance each other. The wave motion is described principally by the equation of continuity. No force term is included (kinematic).

- No applicable to simulate backwater effect
- Applicable for steep slope channel
- Equivalent to uniform flow

Momentum equation in kinematic wave model can be expressed as

$$A = \alpha Q^\beta \quad (c)$$

Differentiating eq. (c) w.r.t. t

$$\frac{\partial A}{\partial t} = \alpha \beta Q^{\beta-1} \frac{\partial Q}{\partial t} \quad (d)$$

From eq. (a) and (d)

$$\frac{\partial Q}{\partial x} + \alpha \beta Q^{\beta-1} \frac{\partial Q}{\partial t} = q$$

This is the kinematic wave model in terms of Q.

α, β from Manning's equation

$$Q = \frac{1}{n} A R^{2/3} S^{1/2} = \frac{1}{n} \frac{A^{5/3}}{P^{2/3}} S^{1/2}$$

$$A = \left(\frac{n P^{2/3}}{\sqrt{S}} \right)^{3/5} Q^{3/5}$$

Comparing to $A = \alpha Q^\beta$

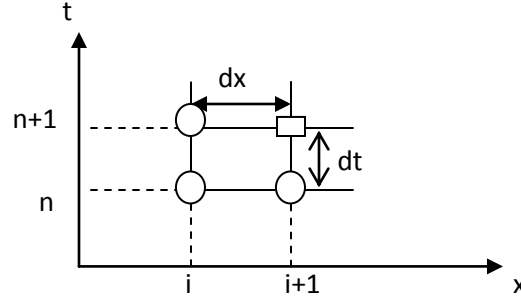
$$\alpha = \left(\frac{n P^{2/3}}{\sqrt{S}} \right)^{0.6} \text{ and } \beta = 0.6$$

5.4 Numerical scheme for kinematic wave model

Finding $Q(x,t)$ at each point on the x - t grid: Q_{i+1}^{n+1}

Given: channel properties, initial and boundary conditions

a. Linear scheme for kinematic wave model



○ : known points, □ : computation point

Discretization

$$\frac{\partial Q}{\partial x} = \frac{Q_{i+1}^{n+1} - Q_i^{n+1}}{\Delta x} : \text{backward difference}$$

$$\frac{\partial Q}{\partial t} = \frac{Q_{i+1}^{n+1} - Q_{i+1}^n}{\Delta t} : \text{backward difference}$$

Non-derivative terms

$$Q = \frac{Q_{i+1}^n + Q_i^{n+1}}{2} : \text{average along diagonal for linearity}$$

$$q = \frac{q_{i+1}^{n+1} + q_{i+1}^n}{2} : \text{average on } i+1 \text{ distance line}$$

Substituting above values in $\frac{\partial Q}{\partial x} + \alpha\beta Q^{\beta-1} \frac{\partial Q}{\partial t} = q$

$$\frac{Q_{i+1}^{n+1} - Q_i^{n+1}}{\Delta x} + \alpha\beta \left(\frac{Q_{i+1}^n + Q_i^{n+1}}{2} \right)^{\beta-1} \frac{Q_{i+1}^{n+1} - Q_{i+1}^n}{\Delta t} = \frac{q_{i+1}^{n+1} + q_{i+1}^n}{2}$$

$$Q_{i+1}^{n+1} \left(\frac{\Delta t}{\Delta x} + \alpha\beta \left(\frac{Q_{i+1}^n + Q_i^{n+1}}{2} \right)^{\beta-1} \right) = \frac{\Delta t}{\Delta x} Q_i^{n+1} + \alpha\beta Q_{i+1}^n \left(\frac{Q_{i+1}^n + Q_i^{n+1}}{2} \right)^{\beta-1} + \left(\frac{q_{i+1}^{n+1} + q_{i+1}^n}{2} \right) \Delta t$$

$$Q_{i+1}^{n+1} = \frac{\frac{\Delta t}{\Delta x} Q_i^{n+1} + \alpha\beta Q_{i+1}^n \left(\frac{Q_{i+1}^n + Q_i^{n+1}}{2} \right)^{\beta-1} + \left(\frac{q_{i+1}^{n+1} + q_{i+1}^n}{2} \right) \Delta t}{\left(\frac{\Delta t}{\Delta x} + \alpha\beta \left(\frac{Q_{i+1}^n + Q_i^{n+1}}{2} \right)^{\beta-1} \right)} \quad (a)$$

Solution technique for linear kinematic wave model for channel routing

1. Supply initial condition at time $t = 0$ on time line $n=1$.
2. Apply Courant condition to each grid point on time line and choose the smallest value as Δt .
3. Advance to next time step, $t = t + \Delta t$ and $n=n+1$.
 - 3.1 Apply upstream boundary conditions (flow hydrograph) to obtain Q_1^{n+1} at $i=1$.

- 3.2 Move to next point on time line $n+1$ by Δx ($x = x + \Delta x$).
- 3.3 Solve for Q_{i+1}^{n+1} using eq. (a).
- 3.4 Repeat steps 3.2-3.3 for all points on the grid
4. Repeat steps 3.1-3.4 for the remaining time periods.

A necessary condition for explicit scheme to be stable is the Courant condition.

Courant no. ≤ 1

$$\text{Courant no.} = \frac{V + |c|}{\Delta x / \Delta t}$$

V = velocity, c = celerity of wave

For computational purposes in an explicit scheme, Δx is specified and kept fixed throughout the computations, while Δt is determined at each time step. Δt at each point on a time line is computed satisfying Courant condition and the smallest one is used.

Stability of kinematic wave

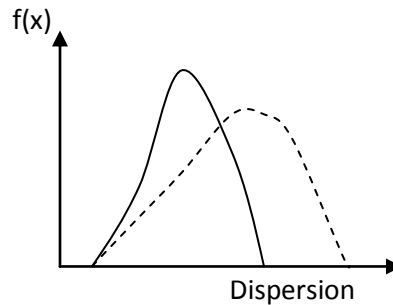
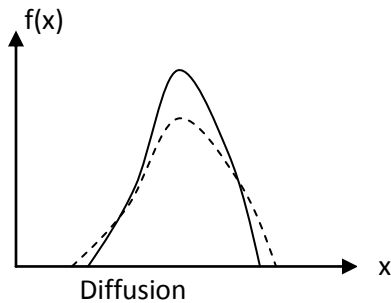
Courant condition: $\Delta t \leq \frac{\Delta x}{c_k}$ where c_k = kinematic wave celerity

Apply Courant condition to each grid point on time line and choose the smallest.

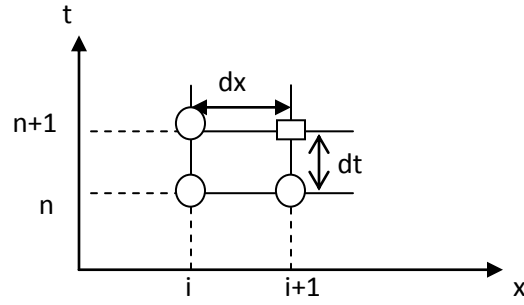
Diffusion and Dispersion

Diffusion: Numerical damping caused by the truncation error is called diffusion. It is related to amplitude error, in which the peak computed by numerical scheme reduces but the timing of peak is same and the shape of function becomes wide.

Error in timing of peak in the numerical solution is called dispersion. It is related to phase error, in which timing of the peak computed by numerical scheme is shifted from the real solution.



b. Non-linear scheme for kinematic wave model



○ : known/given points, □ : computation point

$$\frac{\partial Q}{\partial x} = \frac{Q_{i+1}^{n+1} - Q_i^{n+1}}{\Delta x} : \text{backward}$$

$$\frac{\partial A}{\partial t} = \frac{A_{i+1}^{n+1} - A_{i+1}^n}{\Delta t} : \text{backward}$$

Non-derivative terms

$$q = \frac{q_{i+1}^{n+1} + q_{i+1}^n}{2} : \text{average on } i+1 \text{ distance line}$$

Substituting in continuity equation

$$\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = q$$

$$\frac{Q_{i+1}^{n+1} - Q_i^{n+1}}{\Delta x} + \frac{A_{i+1}^{n+1} - A_{i+1}^n}{\Delta t} = \frac{q_{i+1}^{n+1} + q_{i+1}^n}{2}$$

With $A = \alpha Q^\beta$

$$A_{i+1}^{n+1} = \alpha (Q_{i+1}^{n+1})^\beta \text{ and } A_{i+1}^n = \alpha (Q_{i+1}^n)^\beta$$

Substituting

$$\frac{Q_{i+1}^{n+1} - Q_i^{n+1}}{\Delta x} + \frac{\alpha (Q_{i+1}^{n+1})^\beta - \alpha (Q_{i+1}^n)^\beta}{\Delta t} = \frac{q_{i+1}^{n+1} + q_{i+1}^n}{2}$$

Solving for Q_{i+1}^{n+1}

$$\frac{\Delta t}{\Delta x} Q_{i+1}^{n+1} + \alpha (Q_{i+1}^{n+1})^\beta = \frac{\Delta t}{\Delta x} Q_i^{n+1} + \alpha (Q_{i+1}^n)^\beta + \Delta t \left(\frac{q_{i+1}^{n+1} + q_{i+1}^n}{2} \right) \quad (a)$$

It is non-linear in Q_{i+1}^{n+1} . The above equation can be solved by iteration technique, such as Newton-Raphson method.

Let RHS = C which is known.

$$\text{Residual error is } f(Q_{i+1}^{n+1}) = \frac{\Delta t}{\Delta x} Q_{i+1}^{n+1} + \alpha (Q_{i+1}^{n+1})^\beta - c \quad (b)$$

$$f'(Q_{i+1}^{n+1}) = \frac{\Delta t}{\Delta x} + \alpha \beta (Q_{i+1}^{n+1})^{\beta-1} \quad (c)$$

The objective is to find Q_{i+1}^{n+1} that forces $f(Q_{i+1}^{n+1})$ to become zero.

Newton-Raphson formula

$$(Q_{i+1}^{n+1})_{k+1} = (Q_{i+1}^{n+1})_k - \frac{f(Q_{i+1}^{n+1})_k}{f'(Q_{i+1}^{n+1})_k} \quad (d)$$

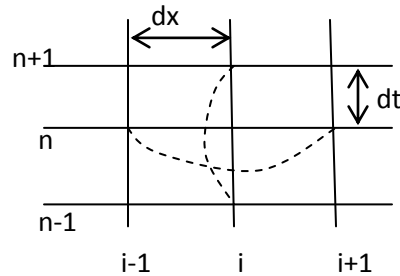
Use linear scheme, compute Q_{i+1}^{n+1} and use it as an initial value.

Solution technique for non-linear kinematic wave model for channel routing

1. Supply initial condition at time $t = 0$ on time line $n=1$.
2. Advance to next time step, $t = t + \Delta t$ and $n=n+1$.
 - 2.1 Apply upstream boundary conditions (flow hydrograph) to obtain Q_1^{n+1} at $i=1$.
 - 2.2 Move to next point on time line $n+1$ by Δx ($x = x + \Delta x$).
 - 2.3 Solve for initial estimate of Q_{i+1}^{n+1} at $k=1$ using linear scheme. Find $f(Q_{i+1}^{n+1})_k$ using eq. (b).
 - 2.4 Find $f'(Q_{i+1}^{n+1})_k$ from eq. (c).
 - 2.5 Find $(Q_{i+1}^{n+1})_{k+1}$ using eq. (b).
 - 2.6 If $f(Q_{i+1}^{n+1})_{k+1} \leq \epsilon$, move to next grid point. Else repeat steps 2.4-2.5 until the condition is satisfied.
 - 2.7 Repeat steps 2.2-2.6 until the last grid point is reached.
3. Repeat steps 2.1-2.7 for the remaining time periods.

5.5 Finite difference scheme for Saint Venant equations (Dynamic wave model)

a. Example of explicit scheme: Leap-frog scheme



Central difference for space and time derivatives

$$\frac{\partial f}{\partial x} = \frac{f_{i+1}^n - f_{i-1}^n}{2\Delta X}$$

$$\frac{\partial f}{\partial t} = \frac{f_i^{n+1} - f_i^{n-1}}{2\Delta t}$$

Non-derivative term

$$f = f_i^n \text{ or } \frac{f_{i-1}^n + f_{i+1}^n}{2} \text{ or } \frac{f_{i-1}^n + f_i^n + f_{i+1}^n}{3}$$

Continuity equation

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = 0$$

$$\frac{A_i^{n+1} - A_i^{n-1}}{2\Delta t} + \frac{Q_{i+1}^n - Q_{i-1}^n}{2\Delta X} = 0$$

$$A_i^{n+1} = A_i^{n-1} - \Delta t \frac{Q_{i+1}^n - Q_{i-1}^n}{\Delta X} \quad (I)$$

Momentum equation

$$\frac{\partial Q}{\partial t} + \frac{\partial(Q^2/A)}{\partial x} + gA \frac{\partial y}{\partial x} - gA(S_0 - S_f) = 0$$

$$\frac{Q_i^{n+1} - Q_i^{n-1}}{2\Delta t} + \frac{1}{2\Delta x} [(Q^2/A)_{i+1}^n - (Q^2/A)_{i-1}^n] + gA_i^n \frac{y_{i+1}^n - y_{i-1}^n}{2\Delta x} - gA_i^n (S_0 - S_f)_i^n = 0$$

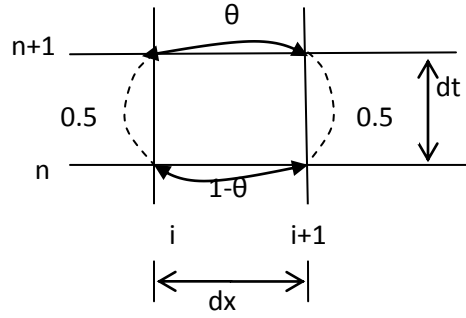
$$Q_i^{n+1} = Q_i^{n-1} - 2\Delta t \left[\frac{1}{2\Delta x} \{ (Q^2/A)_{i+1}^n - (Q^2/A)_{i-1}^n \} + gA_i^n \frac{y_{i+1}^n - y_{i-1}^n}{2\Delta x} - gA_i^n (S_0 - S_f)_i^n \right] \quad (II)$$

b. Example of implicit scheme

(Implicit dynamic wave model)

Implicit four point scheme (Preissman scheme)

In Preissmann scheme, four points are used to evaluate the derivatives. The scheme is forward difference scheme with weight.



Space derivative: $\frac{\partial f}{\partial x} = \theta \frac{(f_{i+1}^{n+1} - f_i^{n+1})}{\Delta x} + (1 - \theta) \frac{(f_{i+1}^n - f_i^n)}{\Delta x}$

Time derivative: $\frac{\partial f}{\partial t} = \frac{1}{2} \left(\frac{f_{i+1}^{n+1} - f_{i+1}^n}{\Delta t} + \frac{f_i^{n+1} - f_i^n}{\Delta t} \right)$

Non-derivative term: $f = \frac{\theta}{2} (f_{i+1}^{n+1} + f_i^{n+1}) + \frac{(1-\theta)}{2} (f_{i+1}^n + f_i^n)$

$\theta = 0.5$: box scheme

$\theta = 0$: fully explicit

$\theta = 1$: fully implicit

For implicit scheme, θ should be between 0.5 to 1.

For θ in between 0.5 to 1, the scheme is unconditionally stable.

Continuity equation

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = 0$$

$$\frac{1}{2} \left(\frac{A_{i+1}^{n+1} - A_{i+1}^n}{\Delta t} + \frac{A_i^{n+1} - A_i^n}{\Delta t} \right) + \theta \frac{(Q_{i+1}^{n+1} - Q_i^{n+1})}{\Delta x} + (1 - \theta) \frac{(Q_{i+1}^n - Q_i^n)}{\Delta x} = 0 \quad (I)$$

Momentum equation

$$\frac{\partial Q}{\partial t} + \frac{\partial(Q^2/A)}{\partial x} + gA \frac{\partial y}{\partial x} - gA(S_0 - S_f) = 0$$

$$\begin{aligned}
& \frac{1}{2} \left(\frac{Q_{i+1}^{n+1} - Q_{i+1}^n}{\Delta t} + \frac{Q_i^{n+1} - Q_i^n}{\Delta t} \right) + \theta \frac{\{(Q^2/A)_{i+1}^{n+1} - (Q^2/A)_i^{n+1}\}}{\Delta X} + (1-\theta) \frac{\{(Q^2/A)_{i+1}^n - (Q^2/A)_i^n\}}{\Delta X} \\
& + g \left\{ \frac{\theta}{2} (A_{i+1}^{n+1} + A_i^{n+1}) + \frac{(1-\theta)}{2} (A_{i+1}^n + A_i^n) \right\} \left\{ \theta \frac{(y_{i+1}^{n+1} - y_i^{n+1})}{\Delta X} + (1-\theta) \frac{(y_{i+1}^n - y_i^n)}{\Delta X} \right\} \\
& - g \left\{ \frac{\theta}{2} (A_{i+1}^{n+1} + A_i^{n+1}) + \frac{(1-\theta)}{2} (A_{i+1}^n + A_i^n) \right\} \left\{ \frac{\theta}{2} ((S_0 - S_f)_{i+1}^{n+1} + (S_0 - S_f)_i^{n+1}) + \frac{(1-\theta)}{2} ((S_0 - S_f)_{i+1}^n + (S_0 - S_f)_i^n) \right\} = 0
\end{aligned} \tag{II}$$

The unknown quantities in implicit finite difference equations are Q_i^{n+1} and y_i^{n+1} . (The terms Q_{i+1}^{n+1} and A_{i+1}^{n+1} should also be computed simultaneously). The area A_{i+1}^{n+1} and the friction slope $(S_f)_i^{n+1}$ can be expressed in terms of Q_i^{n+1} and y_i^{n+1} . All the other terms are known either from the initial conditions or from the previous time step computations.

For $i = 1, 2, \dots, N$, there are $N-1$ reaches. For each reach, 2 equations (continuity and momentum) can be written. So, there are $2N-2$ equations. Additional 2 equations are provided from boundary conditions. $f(x)=0$

Solution technique of resulting equations: Newton-Raphson iteration technique (forcing the residuals to become zero)

Start with values of unknowns from initial condition, previous time step or from extrapolation.

Compute residuals (0-value of $f(x)$).

Find the corrections. (Matrix inversion)

Apply the correction to previous value. Repeat the procedure until two values of corrections do not change significantly.

Value at next time step = value at previous time step + correction

Chapter 6: Method of Characteristics (MOC)

6.1 Introduction

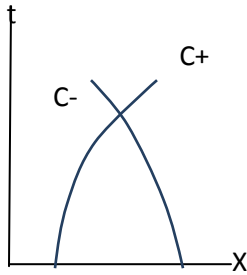
Method of Characteristics (MOC) is a technique of solving partial differential equations (PDE) by converting them to ordinary differential equations (ODE) and solving the ODES by numerical or graphical technique.

Characteristics

Characteristics are curves for plot of $\frac{dx}{dt} = V \pm c$ in x-t plane, where V = velocity of flow and c = celerity of wave. The term $V \pm c$ represents the velocity of propagation of disturbance.

Slope of curve, $\tan\theta = \frac{dt}{dx} = \frac{1}{V \pm c}$

Slope of characteristic line: $\frac{1}{V+c}$ along C+ and $\frac{1}{V-c}$ along C-



Requirements for solution

- Initial condition: e.g. value of variables at $t = 0$
- Boundary conditions (BC): conditions at the boundary of physical system

6.2 Application of MOC to unsteady pipe flow problems

Two basic equations of unsteady pipe flow are as follows.

$$\text{Momentum equation: } L_1 = \frac{\partial V}{\partial t} + V \frac{\partial V}{\partial x} + \frac{1}{\rho} \frac{\partial P}{\partial x} + g \sin\theta + \frac{fV|V|}{2D} = 0 \quad (I)$$

The momentum

$$\text{Continuity equation: } L_2 = \frac{\partial P}{\partial t} + V \frac{\partial P}{\partial x} + \rho c^2 \frac{\partial V}{\partial x} = 0 \quad (II)$$

Where V = Average velocity over the section, ρ = density of fluid, P = pressure at a point, c = celerity of wave, g = acceleration due to gravity, $\sin\theta$ = slope of pipe, f = friction factor

$$c = \sqrt{\frac{K}{\rho \left(1 + \frac{DK}{Et}\right)}} \text{ where } K = \text{Bulk Modulus of fluid, } D = \text{diameter of pipe, } E = \text{Young's modulus of pipe}$$

material, t = thickness of pipe

The unknown variables to compute are V and P.

Combining two equations as $L = L_1 + \lambda L_2 = 0$ where $\lambda = \text{multiplier}$

$$L = \left[\frac{\partial V}{\partial x} (V + \lambda \rho c^2) + \frac{\partial V}{\partial t} \right] + \lambda \left[\frac{\partial P}{\partial x} \left(\frac{1}{\rho \lambda} + V \right) + \frac{\partial P}{\partial t} \right] + g \sin \theta + \frac{fV|V|}{2D} = 0 \quad (\text{III})$$

We have

$$\frac{dV}{dt} = \frac{\partial V}{\partial x} \frac{dx}{dt} + \frac{\partial V}{\partial t} \text{ and } \frac{dP}{dt} = \frac{\partial P}{\partial x} \frac{dx}{dt} + \frac{\partial P}{\partial t}$$

Substituting in the combined expression

$$L = \frac{dV}{dt} + \lambda \frac{dP}{dt} + g \sin \theta + \frac{fV|V|}{2D} = 0 \quad (\text{IV})$$

The above expression is valid

If $\frac{dx}{dt} = (V + \lambda \rho c^2)$ for velocity part and $\frac{dx}{dt} = \left(\frac{1}{\rho \lambda} + V \right)$ for pressure part

Equating

$$(V + \lambda \rho c^2) = \left(\frac{1}{\rho \lambda} + V \right)$$

$$\lambda = \pm \frac{1}{\rho c}$$

Substituting in Eq. IV

$$L = \frac{dV}{dt} \pm \frac{1}{\rho c} \frac{dP}{dt} + g \sin \theta + \frac{fV|V|}{2D} = 0 \quad (\text{V})$$

With $\lambda = \pm \frac{1}{\rho c}$, dx/dt becomes

$$\frac{dx}{dt} = V \pm c \quad (\text{VI})$$

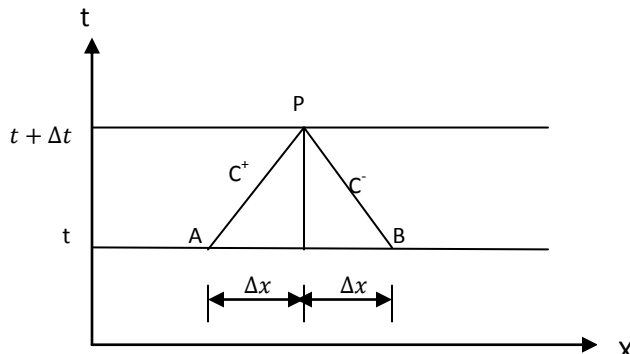
Equation (V) is valid if equation (VI) is satisfied.

Equations (V) and (VI) are the characteristics equations in ODE, which can be solved by FDM.

$V \frac{\partial V}{\partial x}$ in eq. (I) and $V \frac{\partial P}{\partial x}$ in eq. (II) are much smaller than $\frac{\partial V}{\partial t}$ and $\frac{\partial P}{\partial t}$. So, they can also be neglected and

$$\frac{dx}{dt} = \pm c.$$

6.2.1 Finite difference solution of characteristics equation



Given or known: Velocity and pressure at previous time t , i.e. V_A, P_A, x_A, t_A at A and V_B, P_B, x_B, t_B at B

Slope of line AP = $\frac{dt}{dx} = \frac{1}{c}$ and Slope of line BP = $\frac{dt}{dx} = -\frac{1}{c}$

P: Point of intersection of AP and BP represents the solution at P.

Taking C+

$$\frac{dV}{dt} + \frac{1}{\rho c} \frac{dP}{dt} + g \sin \theta + \frac{fV|V|}{2D} = 0$$

Multiplying throughout by $\rho c dt$ and integrating from A to P

$$\rho c \int_A^P dV + \int_A^P dP + \int_A^P \rho c g \sin \theta dt + \int_A^P \rho c dt \frac{fV|V|}{2D} = 0$$

As $\frac{dx}{dt} = c$, replace cdt by dx .

$$\rho c \int_A^P dV + \int_A^P dP + \int_A^P \rho g \sin \theta dx + \int_A^P \rho dx \frac{fV|V|}{2D} = 0$$

Writing in Finite difference form

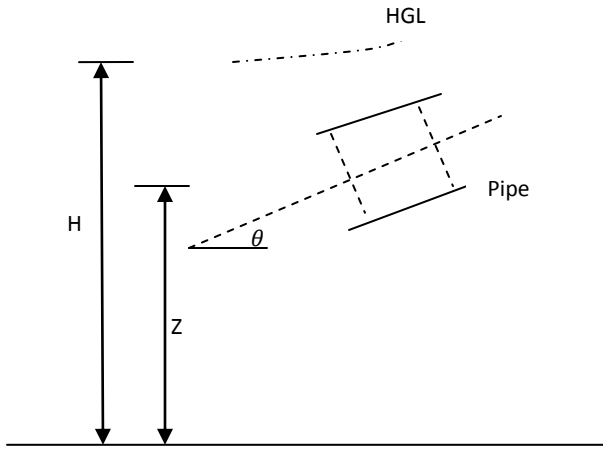
$$\rho c(V_P - V_A) + (P_P - P_A) + \rho g \sin \theta \Delta x + \frac{\rho \Delta x f V_A |V_A|}{2D} = 0 \quad (\text{VII})$$

Similarly, for C- (for line PB)

$$\rho c(V_P - V_B) - (P_P - P_B) + \rho g \sin \theta \Delta x + \frac{\rho \Delta x f V_B |V_B|}{2D} = 0 \quad (\text{VIII})$$

Equations (VII) and (VIII) are solved simultaneously to obtain V_P and P_P .

Equations (VII) and (VIII) can also be expressed in terms of hydraulic gradient line (H) and discharge (Q) in place of P and V.



$$P = \rho g(H - Z)$$

Applying this equation at P and A

$$P_P = \rho g(H_P - Z_P) \text{ and } P_A = \rho g(H_A - Z_A)$$

Subtracting

$$P_P - P_A = \rho g(H_P - H_A) - \rho g(Z_P - Z_A) = \rho g(H_P - H_A) - \rho g \sin \theta \Delta x$$

Substituting $P_P - P_A$ in eq. (VII) and replacing V by $V=Q/A$

$$\frac{\rho c}{A}(Q_P - Q_A) + \rho g(H_P - H_A) - \rho g \sin \theta \Delta x + \rho g \sin \theta \Delta x + \frac{\rho \Delta x f Q_A |Q_A|}{2DA^2} = 0$$

$$H_P = H_A - \frac{c}{gA}(Q_P - Q_A) - \frac{\Delta x f Q_A |Q_A|}{2gDA^2}$$

$$\text{Taking } B = \frac{c}{gA} \text{ and } R = \frac{f \Delta x}{2gDA^2}$$

Then for C+

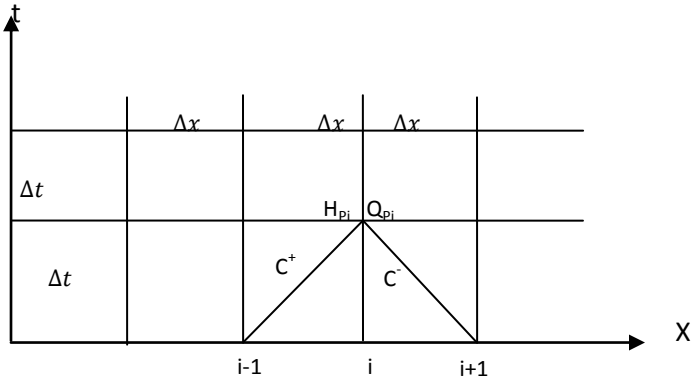
$$H_P = H_A - B(Q_P - Q_A) - RQ_A |Q_A| \quad (\text{IX})$$

Similarly for C-

$$H_P = H_B + B(Q_P - Q_B) + RQ_B |Q_B| \quad (\text{X})$$

Digital Computer solution

Application of eq. (IX) and (X) to compute values of H and Q at grid point i using rectangular grid



Take $\Delta x = L/N$ and $\Delta t = \Delta x/c$

L = Length of pipe, N = No. of reaches

Form the rectangular grids in such a way that C+ and C- are the diagonals of the grid.

Known: Q_{i-1} , H_{i-1} , Q_{i+1} , H_{i+1}

To compute: H_{pi} , Q_{pi}

From eq. (IX) (replacing A by i-1)

$$H_{pi} = H_{i-1} - B(Q_{pi} - Q_{i-1}) - RQ_{i-1}|Q_{i-1}|$$

$$H_{pi} + BQ_{pi} = H_{i-1} + Q_{i-1}(B - R|Q_{i-1}|)$$

$$H_{pi} + BQ_{pi} = C_P \text{ along } C+ \quad (XI)$$

where $C_P = H_{i-1} + Q_{i-1}(B - R|Q_{i-1}|)$ = known term

From eq. (X) (replacing B by i+1)

$$H_{pi} = H_{i+1} + B(Q_{pi} - Q_{i+1}) + RQ_{i+1}|Q_{i+1}|$$

$$H_{pi} - BQ_{pi} = H_{i+1} - Q_{i+1}(B - R|Q_{i+1}|)$$

$$H_{pi} - BQ_{pi} = C_M \text{ along } C- \quad (XII)$$

where $C_M = H_{i+1} - Q_{i+1}(B - R|Q_{i+1}|)$ = known term

Solving eq. (XI) and (XII)

$$H_{pi} = \frac{1}{2}(C_P + C_M)$$

$$Q_{pi} = \frac{1}{B}(C_P - H_{pi})$$

Procedure

Take $\Delta x = L/N$ and $\Delta t = \Delta x/c$.

Compute B and R. $\left(B = \frac{c}{gA}, R = \frac{f\Delta x}{2gDA^2}\right)$

Known: Q_{i-1} , H_{i-1} , Q_{i+1} , H_{i+1} from initial condition or from previous time steps's computation.

For grid point i

Compute C_P and C_M . compute H_{pi} and Q_{pi} .

Perform similar computations for all interior nodes.

Advance to next time step.

Boundary conditions (BC)



Upstream end

Downstream end

At u/s end, only C^- line intersects the boundary and at d/s end, only C^+ line meets the boundary. As the intersection of two characteristics give the solution at a point for H and Q , one condition is needed at each boundary.

Types of BC: constant discharge or head (simplest), time varying discharge or head, relationship of discharge and head in algebraic or differential equation form

Chapter 7: Simulation of Groundwater Flow

7.1 Basic equation of groundwater flow and flow net analysis

a. Darcy's law

According to Darcy's law, the velocity through porous media is proportional to hydraulic gradient.

$v = -K \frac{dh}{dl}$ where v = velocity of flow known as Darcy velocity or specific discharge, K = hydraulic conductivity, dh/dl = hydraulic gradient

$$\text{Discharge (Q)} = Av = -KA \frac{dh}{dl}$$

Darcy's equation in terms of transmissivity (T)

$T = KH$ where H = thickness of aquifer

$$v = -TH \frac{dh}{dl}$$

b. Steady continuity equation for incompressible flow

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (I)$$

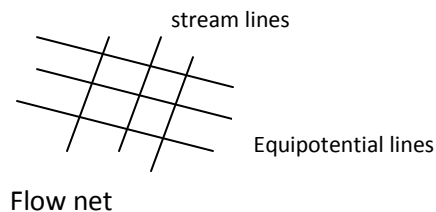
u, v and w : velocity in x, y and z direction

Stream line, equipotential line and flow net

Stream line (flow line) is an imaginary curve drawn through the flow field in such a way that the tangent to it at any point indicates the direction of velocity vector at that point. Along streamline, stream function is constant. In steady flow, streamline coincides with the path line of flow. An equipotential line is a line along which velocity potential (hydraulic head in groundwater flow) is constant. A grid obtained by drawing a series of equipotential lines and stream lines is called a flow net.

Rules for constructing flow nets in isotropic medium

- Streamlines are drawn in the direction of flow; they can be curvilinear. They form a set of streamtubes. Streamlines cannot intersect.
- Equipotential lines are drawn perpendicular to the streamlines



Total energy head of flow potential at any point on the water table is $\frac{P}{\gamma} + z = h$ = piezometric head.

Taking atmospheric pressure as datum, $h = Z$. Therefore, under steady conditions elevation at any point on the water table is equal to energy head and flow lines lie perpendicular to water table contours. As contours of h represents ϕ , Darcy's equation can also be written as

$$v = -K \frac{d\phi}{dl} \text{ or } -TH \frac{d\phi}{dl}$$

c. Laplace equation in groundwater flow

Steady continuity equation for incompressible flow

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (I)$$

ϕ = Velocity potential

$$u = -\frac{\partial \phi}{\partial x}, v = -\frac{\partial \phi}{\partial y}, w = -\frac{\partial \phi}{\partial z}$$

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \quad (i)$$

This is Laplace equation for velocity potential.

For homogeneous and isotropic formation

$$u = -K \frac{\partial h}{\partial x}, v = -K \frac{\partial h}{\partial y}, w = -K \frac{\partial h}{\partial z}$$

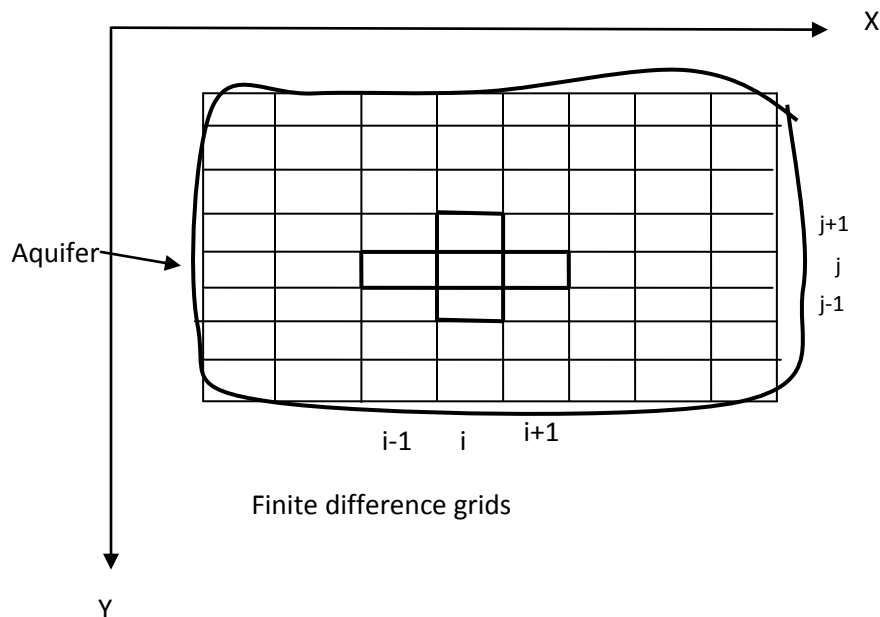
Substituting u, v and w in eq. (i)

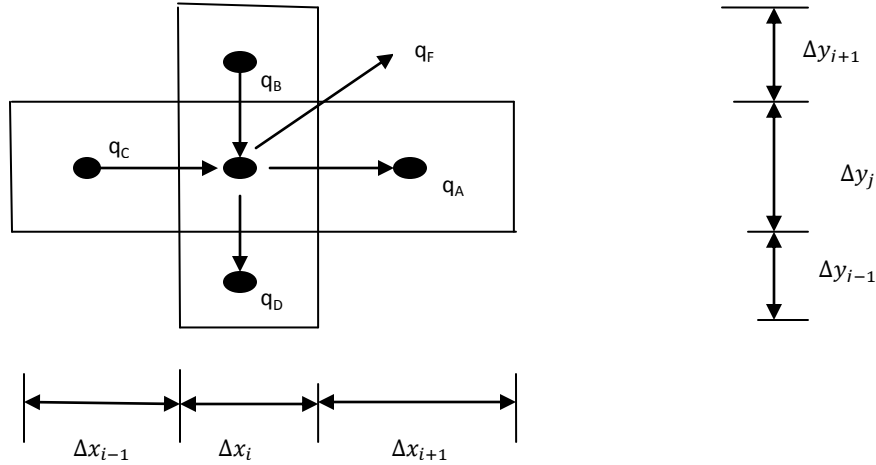
$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} + \frac{\partial^2 h}{\partial z^2} = 0 \text{ . This is the Laplace equation for groundwater head.}$$

For 2D flow,

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} = 0$$

7.2 Finite difference scheme for 2D groundwater simulation





Enlarged view of interior grids defined by $i-1, i, i+1$ and $j-1, j$ and $j+1$

Flow takes place from center to center of grid.

q_A, q_B, q_C, q_D = grid to grid flow rates per unit area (Darcy flux)

q_F = net groundwater withdrawal rate per unit area from grid (i,j)

Continuity equation for grid (i, j)

inflow-outflow = rate of change of storage

$$\Delta x_i \Delta y_j [q_C - q_D - q_A + q_B - q_F] = S \frac{\partial \phi}{\partial t} \quad (I)$$

Where S = storage coefficient, ϕ = potential head, t = time

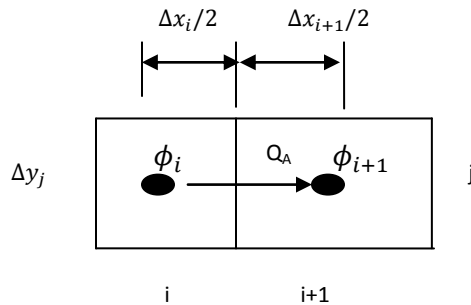
Darcy's equation

$$Q = -KA \frac{\partial \phi}{\partial x}$$

In terms of transmissivity

$$Q = -TH \frac{\partial \phi}{\partial x} \quad (II)$$

Consider (i,j) and $(i+1,j)$ grid



Finite difference form of Darcy's equation (II), considering isotropic aquifer

$$Q_A = -T(\Delta y_j) \frac{\phi_{i+1,j} - \phi_{i,j}}{\frac{\Delta x_{i+1} + \Delta x_i}{2}} = -2T(\Delta y_j) \frac{\phi_{i+1,j} - \phi_{i,j}}{\Delta x_{i+1} + \Delta x_i} \quad (III)$$

(For anisotropic medium K is different in x and Y directions, So weighted average transmissivity of two grids in X and Y direction is taken.)

$$q_A = -\frac{Q_A}{\Delta x_i \Delta y_j} = A_{i,j}(\phi_{i+1,j} - \phi_{i,j})$$

$$\text{Where } A_{i,j} = \frac{2T}{\Delta x_i(\Delta x_i + \Delta x_{i+1})}$$

Similarly, other Darcy fluxes in finite difference form are

$$q_C = C_{i,j}(\phi_{i-1,j} - \phi_{i,j})$$

$$q_D = D_{i,j}(\phi_{i,j+1} - \phi_{i,j})$$

$$q_B = B_{i,j}(\phi_{i,j-1} - \phi_{i,j})$$

Where

$$B_{i,j} = \frac{2T}{\Delta y_j(\Delta y_{j-1} + \Delta y_j)}$$

$$C_{i,j} = \frac{2T}{\Delta x_i(\Delta x_i + \Delta x_{i-1})}$$

$$D_{i,j} = \frac{2T}{\Delta y_j(\Delta y_{j+1} + \Delta y_j)}$$

Substituting values of Darcy fluxes in eq. (I)

$$C_{i,j}(\phi_{i-1,j} - \phi_{i,j}) + D_{i,j}(\phi_{i,j+1} - \phi_{i,j}) + A_{i,j}(\phi_{i+1,j} - \phi_{i,j}) + B_{i,j}(\phi_{i,j-1} - \phi_{i,j}) - q_F = S \frac{\partial \phi}{\partial t} \quad (V)$$

Dropping the subscripts of coefficients

$$A\phi_{i+1,j} + B\phi_{i,j-1} + C\phi_{i-1,j} + D\phi_{i,j+1} - E\phi_{i,j} = S \frac{\partial \phi}{\partial t} + q_F \quad (VI)$$

Where $E = A+B+C+D$

If $\Delta x = \Delta y$, then the continuity equation reduces to

$$\phi_{i+1,j} + \phi_{i,j-1} + \phi_{i-1,j} + \phi_{i,j+1} - 4\phi_{i,j} = \frac{\Delta x^2}{T} \left(S \frac{\partial \phi}{\partial t} + q_F \right)$$

For steady flow with $q_F = 0$

$$\phi_{i+1,j} + \phi_{i,j-1} + \phi_{i-1,j} + \phi_{i,j+1} - 4\phi_{i,j} = 0$$

$\phi_{i,j}$ is equal to average of four neighbouring grid values.

At boundaries, four grids will not be available. So, the boundary condition should be taken into account to compute coefficients.

For impervious boundary on all sides, the coefficient from that side is zero. $j=1$, $B=0$ and $j=n$, $D=0$

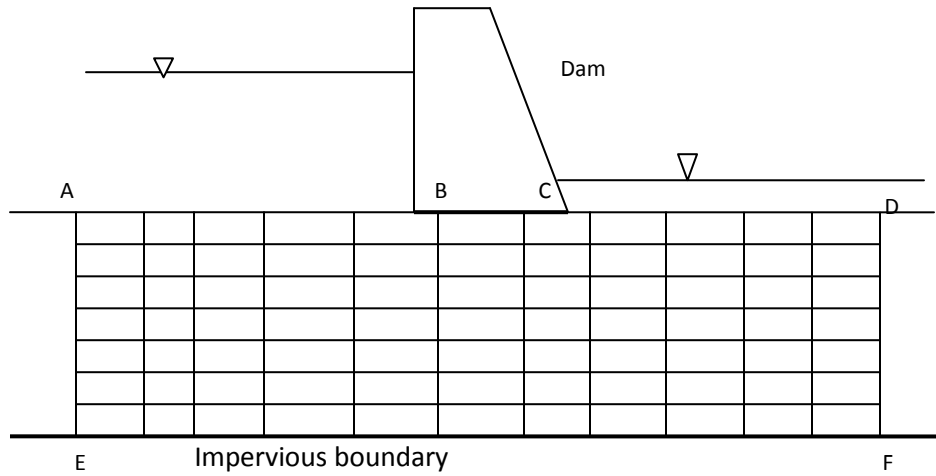
$i=1$, $C=0$ and $i=n$, $A=0$

For grid elements adjacent to a river, only one value of space increment is needed.

$$\text{e.g. } A = \frac{2T}{\Delta x^2}$$

After computing ϕ at different grids, equipotential lines are drawn by interpolation and streamlines are drawn perpendicular to these lines.

7.3 Simulation of seepage under a dam based on steady state 2D model



Finite difference grid for seepage under dam

From eq. (VI), the continuity equation is

$$A\phi_{i+1,j} + B\phi_{i,j-1} + C\phi_{i-1,j} + D\phi_{i,j+1} - E\phi_{i,j} = S \frac{\partial \phi}{\partial t} + q_F$$

For steady state condition with no withdrawal

$$A\phi_{i+1,j} + B\phi_{i,j-1} + C\phi_{i-1,j} + D\phi_{i,j+1} - E\phi_{i,j} = 0$$

$$\phi_{i,j}^+ = (A\phi_{i+1,j} + B\phi_{i,j-1} + C\phi_{i-1,j} + D\phi_{i,j+1})/E$$

Where +: present iteration, without +: previous iteration

Iterative procedure to compute potential at each grid and seepage rate

1. Consider the boundary condition for evaluating the continuity equation at boundary.

In the figure, BC and EF are impervious boundaries, AB and CD are equipotential head (constant head). AE and DF can also be considered as barrier boundary if the grids extend far enough so that no flow occurs across AE and DF.

2. Assign initial value of ϕ at each grid. (usually 0 or estimated value)

3. Compute $\phi_{i,j}^+$.

4. Compute $e = |\phi_{i,j}^+ - \phi_{i,j}|$

5. If e is less than some specified value, stop the iteration.

Else Repeat steps 3 to 4 until the condition is satisfied.

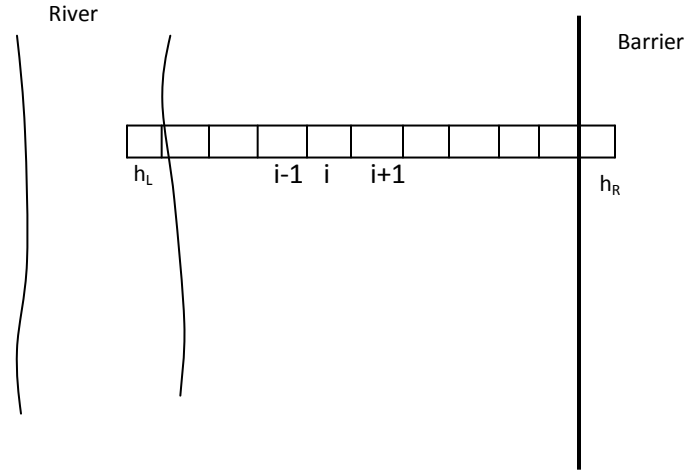
6. Compute seepage rate by using Darcy equation.

Horizontal seepage from grid $i-1$ is $Q_{i,j}^H = C_{i,j} \Delta x_i \Delta y_j (\phi_{i-1,j} - \phi_{i,j})$

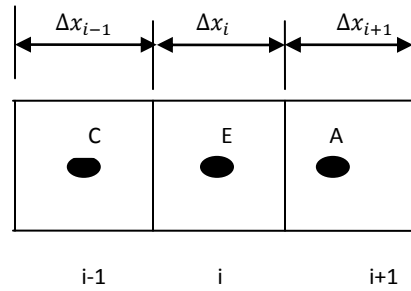
Vertical seepage from grid $j-1$ is $Q_{i,j}^V = B_{i,j} \Delta x_i \Delta y_j (\phi_{i,j-1} - \phi_{i,j})$

7.4 One Dimensional implicit model for simulating river stage-water table interactions

(River recharging groundwater aquifer or groundwater aquifer supplying water to river)



1D finite difference grid



Partial differential equation for 1D unsteady groundwater flow is

$$\frac{\partial}{\partial x} \left(T \frac{\partial \phi}{\partial x} \right) = S \frac{\partial \phi}{\partial t} + q_F$$

Where T = Transmissivity, ϕ = water table elevation for an unconfined aquifer or the elevation of piezometric surface for confined aquifer, S = storage coefficient, q_F = net groundwater withdrawal per unit area

From eq. (V), the continuity equation for grid i is

$$A_i(h_{i+1}^+ - h_i^+) + C_i(h_{i-1}^+ - h_i^+) = \frac{S_i}{\Delta t}(h_i^+ - h_i) + q_F$$

Where h and h^+ = water table elevation at the beginning and end of time step Δt .

$$A_i = \frac{2T}{\Delta x_i(\Delta x_i + \Delta x_{i+1})}, C_i = \frac{2T}{\Delta x_i(\Delta x_i + \Delta x_{i-1})}$$

$$\text{Defining } E_i = -\left(A_i + C_i + \frac{S_i}{\Delta t}\right) \text{ and } Q_i = -\frac{S_i}{\Delta t}h_i + q_F$$

The continuity equation becomes

$$C_i h_{i-1}^+ + E_i h_i^+ + A_i h_{i+1}^+ = Q_i$$

Writing this equation for each grid results in tri-diagonal coefficient matrix

$$\begin{bmatrix} E_1 & A_1 & 0 & 0 & 0 & 0 & 0 \\ C_2 & E_2 & A_2 & 0 & 0 & 0 & 0 \\ 0 & C_3 & E_3 & A_3 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & C_n & A_n \end{bmatrix} \begin{bmatrix} h_1^+ \\ h_2^+ \\ h_3^+ \\ \vdots \\ h_n^+ \end{bmatrix} = \begin{bmatrix} Q_1 - C_1 h_L \\ Q_2 \\ Q_3 \\ \vdots \\ Q_n - A_n h_R \end{bmatrix}$$

h_L = river stage at potential boundary

As the right boundary is barrier boundary, $A_n = 0$.

The system of linear equations is solved to find h .

The seepage rate between grid elements is computed by

$$Q_i = C_i \Delta x_i (h_{i-1}^+ - h_i^+)$$

Loss or gain at the river = seepage at grid element 1. ($i=1$)